Module 4 Lab Reflective Journal

Course: ITAI 1371
Assignment: Module 4 Lab Exercise

John Grubbs

I really enjoyed this assignment because it was the first time we wrote our own code rather than just reading and running the provided examples. Creating my own visualizations helped me better understand how the code works and how different plots can reveal patterns in a dataset.

Exploring the Titanic data also helped me see the value of looking beyond summary statistics. Comparing passenger characteristics and survival outcomes encouraged me to ask questions about the data and form my own observations. This assignment gave me more confidence in using Python for data analysis.

Keira Burdine

In this lab, I developed a better understanding of Exploratory Data Analysis (EDA) and why it is an important step before building a machine learning model. Before completing this assignment, I understood that data needed to be reviewed before modeling, but this lab helped me see how much information can be discovered just by examining summary statistics and creating visualizations. EDA is similar to investigating a dataset for clues. It can reveal missing values, differences between groups, unusual values, and relationships that may affect a future machine learning model.

One of the first things I learned was how useful commands such as df.head(), df.info(), and df.describe() are when first exploring a dataset. Using the Titanic dataset, df.info() showed that there were 891 entries and 12 columns. It also helped me identify missing information in columns such as Age and Cabin. This showed me why checking the condition of the data is necessary before trying to train a model. The descriptive statistics gave me additional information, including an average passenger age of approximately 29.7 years, an overall survival rate of about 38.4%, and fares ranging from $0 to over $512. The visualizations helped the information become much easier to understand. The survival count plot showed that more passengers died than survived. When survival was compared by passenger class, there was a noticeable difference between first-class and third-class passengers. First-class passengers appeared to have a much better chance of surviving. Gender was another strong pattern because females had a much higher survival rate than males. The age histograms also showed how survival differed across age groups. These visualizations helped me understand that a graph can reveal relationships that may not be obvious when looking only at averages or counts. The student

experimentation section helped me apply what I learned instead of only running prewritten code. I created a visualization comparing survival by port of embarkation and observed differences between passengers who boarded at Southampton, Cherbourg, and Queenstown. I also created a boxplot comparing fare and survival. These exercises helped reinforce how categorical and continuous variables require different visualization techniques.

A challenge I experienced during the lab was that several sections of the notebook opened with the Python code combined into one long line. This caused syntax errors even though the lab was supposed to be fully coded. I had to examine the error messages, separate the commands onto individual lines, and make sure explanatory text was marked as comments using the # symbol. At first, the errors were frustrating because I expected the provided cells to run automatically. However, correcting the cells gave me additional practice with Python syntax and helped me better understand how Python interprets code.

Overall, this lab changed the way I think about preparing data for machine learning. I now understand that creating a model should not be the first step. I need to explore the data, identify problems, understand the variables, and look for meaningful patterns first. EDA provides the foundation for making better decisions about how the data should be prepared and which features may be important to a model.

Luis Adrian

In this lab I learn about the concept of “Exploratory Data analysis” . I learned that statistics can show averages and counts, while graphs make it easier to notice patterns, differences, and unusual values. Like using an example of the Titanic about analyzing the rate of how many could survive the accident, providing the data, and based on charts and plots.

And I pretty enjoy doing this lab because, I do not only analyze and learn about how each cells works and respond, I have the chance to build my own code based a bit of the previous ones to make a plot or a chart that can explain and give result about what the experiment require that I show in my own cell.

And the most challenging part that I found was to build my own code based on what the experiment expects that I solve correctly, and try to build the code without any syntax error, going from error and solving to finish each code.

Overall, I feel that this lab is gonna be very important to the next activities that we are gonna require for the semester, and I hope that we can get more labs like this one that are more interactive.